

① [12.4.2]

Using simpler notation, we're asked to show:

$$\hat{R}_y(-\epsilon_y) \hat{R}_x(-\epsilon_x) \hat{R}_y(\epsilon_y) \hat{R}_x(\epsilon_x) = \hat{R}_z(-\epsilon_x \epsilon_y)$$

$$\hat{R}_x(\epsilon_x) \Leftrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -\epsilon_x \\ 0 & \epsilon_x & 1 \end{pmatrix} + O(\epsilon^2) ; \hat{R}_y(\epsilon_y) \Leftrightarrow \begin{pmatrix} 1 & 0 & \epsilon_y \\ 0 & 1 & 0 \\ -\epsilon_y & 0 & 1 \end{pmatrix}$$

$$\hat{R}_z(\epsilon_z) \Leftrightarrow \begin{pmatrix} 1 & -\epsilon_z & 0 \\ \epsilon_z & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\hat{R}_y(\epsilon_y) \hat{R}_x(\epsilon_x) \overset{\text{1st order}}{\Leftrightarrow} \begin{pmatrix} 1 & \epsilon_x \epsilon_y & \epsilon_y \\ 0 & 1 & -\epsilon_x \\ -\epsilon_y & \epsilon_x & 1 \end{pmatrix} \equiv \hat{B}$$

$$\hat{R}_y(-\epsilon_y) \hat{R}_x(-\epsilon_x) \Leftrightarrow \begin{pmatrix} 1 & \epsilon_x \epsilon_y & -\epsilon_y \\ 0 & 1 & \epsilon_x \\ \epsilon_y & -\epsilon_x & 1 \end{pmatrix} \equiv \hat{A}$$

$$\hat{A} \hat{B} = \begin{pmatrix} (1+\epsilon_y^2) & \epsilon_x \epsilon_y & -\epsilon_x^2 \epsilon_y \\ -\epsilon_x \epsilon_y & (1+\epsilon_x^2) & 0 \\ 0 & \epsilon_x \epsilon_y^2 & 1+\epsilon_x^2+\epsilon_y^2 \end{pmatrix}$$

Zero terms of order $\epsilon_x^2, \epsilon_y^2$:

$$= \begin{pmatrix} 1 & \epsilon_x \epsilon_y & 0 \\ -\epsilon_x \epsilon_y & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

QED

② [12.5.3]

$$1) \quad \langle \hat{J}_x \rangle = \frac{1}{2} \langle \hat{J}_+ + \hat{J}_- \rangle = \frac{1}{2} [\langle \hat{J}_+ \rangle + \langle \hat{J}_- \rangle]$$

$$= \frac{1}{2} [\langle j m | \hat{J}_+ | j m \rangle + \langle j m | \hat{J}_- | j m \rangle]$$

$$= \frac{1}{2} [C_{jm}^+ \langle j m | j m+1 \rangle + C_{jm}^- \langle j m | j m-1 \rangle] \quad \left(\text{see Eq. 12.5.20 for values of } C_{jm}^\pm \right)$$

→ Note that even if $m = \pm j$ (and either $|j m+1\rangle$ or $|j m-1\rangle$ doesn't exist) then it still vanishes since $\hat{J}_+$ or $\hat{J}_-$ annihilates the state — i.e. $C_{jm}^\pm$ is zero.

$$\text{Similarly, } \langle \hat{J}_y \rangle = \frac{1}{2i} [\langle \hat{J}_+ \rangle - \langle \hat{J}_- \rangle] = 0.$$

$$2) \quad \langle \hat{J}_x^2 \rangle = \langle \hat{J}^2 - \hat{J}_y^2 - \hat{J}_z^2 \rangle \quad \text{But } \langle \hat{J}_x^2 \rangle = \langle \hat{J}_y^2 \rangle \text{ by symmetry,}$$

$$\text{so } 2 \langle \hat{J}_x^2 \rangle = \langle \hat{J}^2 - \hat{J}_z^2 \rangle$$

$$= \hbar^2 j(j+1) - \hbar^2 m^2$$

$$\langle \hat{J}_x^2 \rangle = \langle \hat{J}_y^2 \rangle = \frac{\hbar^2}{2} [j(j+1) - m^2].$$

$$3) \quad \Delta J_x = \Delta J_y = \sqrt{\frac{\hbar^2}{2} [j(j+1) - m^2]} = \frac{\hbar}{\sqrt{2}} \sqrt{j^2 + j - m^2}$$

$$(\Delta J_x)^2 (\Delta J_y)^2 = \frac{\hbar^4}{4} [j(j+1) - m^2]^2 \geq |\langle \hat{J}_x \hat{J}_y \rangle|^2$$

$$\langle \hat{J}_x \hat{J}_y \rangle = \frac{i}{4} \langle (\hat{J}_+ + \hat{J}_-)(\hat{J}_+ - \hat{J}_-) \rangle$$

$$= \frac{i}{4} \langle \hat{J}_+^2 + \hat{J}_-^2 + [\hat{J}_-, \hat{J}_+] \rangle$$

$$= \frac{i}{4} \langle \hat{J}_+^2 + \hat{J}_-^2 - 2\hbar \hat{J}_z \rangle$$

$$= \frac{i}{4} \left[C_{j, m+1}^+ C_{j, m}^+ \langle j, m | j, m+2 \rangle + C_{j, m-1}^- C_{j, m}^- \langle j, m | j, m-2 \rangle - 2\hbar^2 m \right]$$

$$= \frac{i}{2} \hbar^2 m \quad (\text{Note that } \hat{J}_x \hat{J}_y \text{ is not Hermitian!})$$

$$\text{so } |\langle \hat{J}_x \hat{J}_y \rangle|^2 = \frac{\hbar^4}{4} m^2$$

Back to inequality:

$$\frac{\hbar^4}{4} (j(j+1) - m^2)^2 \geq \frac{\hbar^4}{4} m^2$$

$$j^2(j+1)^2 - 2m^2 j(j+1) + m^4 \geq m^2$$

Recall that $j \geq 0$ and $|m| \leq j$

so $j(j+1) \geq m^2$ and $j^2(j+1)^2 \geq m^4$

4)

If $j = \pm m$, $j^2 = m^2$:

$$j^2(j+1)^2 - 2j^3(j+1) + j^4 \stackrel{?}{=} j^2$$

$$j^4 + 2j^3 + j^2 - j^2 - 2j^4 - 2j^3 + j^4 \stackrel{?}{=} 0 \quad \checkmark$$

③ [12.6.1]

1) No angular dependence: $l=m=0$.

2) Assume $V(r)$ vanishes at large r (as stated).

With $l=0$, radial equation is $\frac{-\hbar^2}{2\mu} \frac{d^2}{dr^2} \psi_E(r) = E \psi_E(r) - \int_0^{\infty} V(r) \psi$

$$\Rightarrow A \left(\frac{-1}{a_0} \right)^2 \left(\frac{-\hbar^2}{2\mu} e^{-r/a_0} \right) = E A e^{-r/a_0}$$

$$\Rightarrow E = \frac{-\hbar^2}{2\mu a_0^2}$$

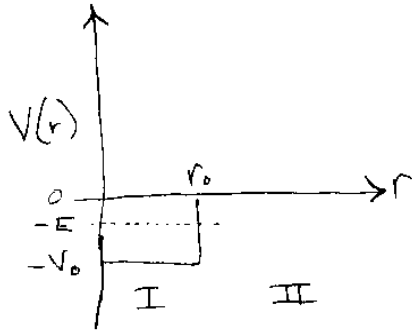
3) Back to radial Schr. eqn:

$$E \psi(r) + \frac{\hbar^2}{2\mu} \left(\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} \right) \psi = V(r) \psi$$

$$\frac{\hbar^2}{2\mu} \cdot \frac{2}{r} \frac{d}{dr} \psi = V(r) \psi = \frac{\hbar^2}{\mu r} \left(\frac{-1}{a_0} \right) \psi$$

$$V(r) = \frac{-\hbar^2}{\mu a_0 r}$$

(4) [12.6.9] — A little Bessel function practice.



Region I: $\psi = j_0(kr) = \frac{\sin k'r}{k'r}$

where $k' = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$

Region II (classically forbidden)

$\psi = \frac{e^{-\kappa r}}{\kappa r}$ where $\kappa = \frac{\sqrt{2mE}}{\hbar}$

Match at r_0 : $\frac{1}{j_0(k'r_0)} \frac{d}{dr} \bigg|_{r_0} j_0(k'r_0) = \frac{-\kappa r}{e^{-\kappa r}} \frac{d}{dr} \bigg|_{r_0} \frac{e^{-\kappa r}}{\kappa r}$

$$\frac{k'r_0}{\sin k'r_0} \left(\frac{k'r_0}{\sin k'r_0} - \frac{\sin k'r_0}{k'r_0^2} \right) = k' \cot k'r_0 - \frac{1}{r_0}$$

$$= \frac{\kappa r_0}{e^{-\kappa r_0}} \left(-\kappa - \frac{1}{r_0} \right) \left(\frac{e^{-\kappa r_0}}{\kappa r_0} \right) = -\kappa - \frac{1}{r_0}$$

$$\text{so } k' \cot k'r_0 - \frac{1}{r_0} = -\kappa - \frac{1}{r_0}$$

$$\frac{k'}{\kappa} = -\tan k'r_0$$

This is the same situation as 5.2.6 except the wave fn must vanish at $r \rightarrow 0$ so "even" solutions from 5.2.6 don't exist. At critical $V_0 = \frac{\pi^2 \hbar^2}{8m r_0^2}$, solution has $E \rightarrow 0$: not bound.